

Substring Order I

Time limit: 1.00 s

Memory limit: 512 MB

You are given a string of length n . If all of its distinct substrings are ordered lexicographically, what is the k th smallest of them?

Input

The first input line has a string of length n that consists of characters a–z.

The second input line has an integer k .

Output

Print the k th smallest distinct substring in lexicographical order.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq k \leq \frac{n(n+1)}{2}$
- It is guaranteed that k does not exceed the number of distinct substrings.

Example

Input	Output
babaacbaab 10	aba

Explanation: The 10 smallest distinct substrings in order are a, aa, aab, aac, aacb, aacba, aacbaa, aacbaab, ab, and aba.